Bachelor of Engineering(B.E.) in
Computer Engineering (BCT)
Electronics, Communication & Information Engg. (BEI)
Third Year Second Part
Elective II Course

Analysis of Algorithms
[Subject Code: ENCT388]
Course Introduction

by:

`Er. Bikal Adhikari

(*B.E., M.Sc. Engg., M.Sc. ISE*)

Assistant Professor, Dept. of Electronics & Computer Engg.

Pulchowk Campus, IOE, TU

COURSE STRUCTURE

ANALYSIS OF ALGORITHMS
ENCT388

Lecture: 3 **Year:** III

Tutorial: 2 **Part:** II

Practical: 1

Course Objectives:

The objective of this course is to provide students with a strong foundation in the analysis of algorithm efficiency and computational complexity. It aims to develop understanding of key algorithm design paradigms, including divide-and-conquer, greedy methods, dynamic programming, and backtracking, and covers fundamental concepts of NP completeness and approximation algorithms.

Chapter 1. Introduction to Algorithm Analysis

5 hours

1.1 Algorithm and its properties, RAM model, time and space complexity, detailed analysis of algorithms, Concept of Aggregate Analysis

1.2 Asymptotic notations (Big-O, Big-Ω and Big-Θ), their geometrical interpretations and examples

1.3 Concept of best case, average case and worst case performance of an algorithm

1.4 Modeling algorithms by recurrence relation

1.5 Solving recurrence relation for evaluating computational complexity

- 1.5.1 Recursion tree method
- 1.5.2 Substitution method
- 1.5.3 Using masters theorem

Chapter 2. Iterative and Numeric Algorithms

8 hours

2.1 Algorithm for GCD and Fibonacci number
2.2 Sequential search
2.3 Review of bubble sort, selection sort, and insertion sort algorithms
2.4 Number theoretic notations
2.5 Euclid's and Extended Euclid's algorithms
2.6 Solving modular linear equations using Chinese remainder theorem
2.7 Fermat's theorem
2.8 Miller-Rabin randomized primility test and algorithm

Chapter 3. Divide and Conquer Algorithms

8 hours

3.1 Binary search, min max finding algorithm

3.2 Analysis of sorting algorithms

- 3.2.1. Merge sort
- 3.2.2. Heap sort
- 3.2.3. Quick sort
- 3.2.4. Randomized Quick sort

3.3. Order Statistics

- 3.3.1. Selection in expected linear time
- 3.3.2. Selection in worst case linear time

Chapter 4. Greedy Algorithms

7 hours

4.1 Basic concepts
4.2 Fractional knapsack problem
4.3 Job sequencing with deadlines
4.4 Analysis of minimum spanning trees related algorithms
4.5 Analysis of single source shortest path algorithm

Unit 5. Dynamic Programming

8 hours

5.1 Basic concepts
5.2 All pair shortest path algorithm
5.3 Travelling salesperson problem
5.4 String editing
5.5 0/1 knapsack problem using dynamic programming
5.6 Matrix chain multiplication
5.7 Flow shop scheduling

Unit 6. Backtracking Techniques

4 hours

6.1 Basic concepts
6.2 The N-Queen problem
6.3 Sum of subsets
6.4 Graph coloring
6.5 Hamiltonian cycles
6.6 0/1 knapsack problem using backtracking approach

Chapter 7. NP-Hard and NP-Complete Problems

5 hours

7.1 Basic concepts
7.2 Cook’s theorem
7.3 NP-Hard graph problems
7.4 NP-Hard scheduling problems
7.5 NP-Hard code generation problems
7.6 Simplified NP-hard problems
7.7 Approximation algorithms: ε- approximation, polynomial time approximation scheme, probabilistically good algorithms
7.8 Vertex cover problem, subset sum problem

TUTORIALS

Tutorial 1. Algorithm analysis
Tutorial 2. Iterative and numeric algorithms
Tutorial 3. Divide and conquer algorithms
Tutorial 4. Greedy algorithms
Tutorial 5. Dynamic programming
Tutorial 6. Backtracking techniques
Tutorial 7. NP-Hard and NP-complete problems

PRACTICAL

1. Implementation and complexity analysis of iterative, numeric and recursive algorithms
2. Implementation and complexity analysis of greedy algorithms
3. Implementation and complexity analysis of algorithms involving divide and conquer strategy
4. Implementation and complexity analysis of algorithms based on dynamic programming
5. Implementation and complexity analysis of algorithms using backtracking concept

Marks Distribution

Chapter	Hours	Marks*
1	5	6
2	8	10
3	8	11
4	7	9
5	8	12
6	4	5
7	5	7
Total	**45**	**60**

REFERENCES

1. Horowitz, E., Sahni, S., Rajasekaran, S. (2007). Fundamentals of computer algorithms. Universities Press.
2. Cormen, T. H., Leiserson, C. E., Rivest, R. L., Stein, C. (2022). Introduction to algorithms. MIT Press.
3. Kleinberg, J., Tardos, É. (2006). Algorithm design. Pearson.
4. Skiena, S. S. (2020). The algorithm design manual. Springer.
5. Dasgupta, S., Papadimitriou, C. H., Vazirani, U. V. (2006). Algorithms. McGraw-Hill Education.









